

Heuristic Evaluation

1. Visibility of System Status
 - Transcripts aren't visible directly
 - Important information isn't visible
 - No error messages
 - System doesn't notify users of changes
2. Match Between System & the Real World
 - Naming isn't accurate and causes confusion
 - Course selection tags don't convey proper information
 - Filtration terms/tags aren't meaningful
3. User Control & Freedom
 - Course acceptance requires deep navigation (hard to undo or return)
 - No notifications of changes → users can't reverse actions
 - Action logging lacks detail to track or undo mistakes
4. Consistency & Standards
 - Naming inconsistencies
 - Different interfaces look very similar (e.g., transfers vs. specialization), causing confusion
5. Error Prevention
 - Information can be overridden without warning
 - System may calculate values incorrectly
 - Error prevention is weak
 - Filtration isn't accurate
 - Course acceptance workflow leads to errors
6. Recognition Rather Than Recall
 - Navigation isn't intuitive
 - Mobile experience isn't intuitive; system should ideally be an app
 - Course selection tags hide extra needed information
 - Filtration doesn't allow multiple selections → users must remember inputs
7. Flexibility & Efficiency of Use
 - Changing student data isn't automated; requires too many steps
 - Attendance warnings are manual
 - Assigning advisors must be done one-by-one
 - Reports must be exported and built manually
 - Student actions aren't easy to perform
 - Filtration doesn't accept multiple inputs
 - New users require training due to inefficient, complex workflows
8. Aesthetic & Minimalist Design
 - Navigation is unintuitive and possibly cluttered
 - User flow is complicated
 - Extra data is hidden inside menus instead of being visible
9. Help Users Recognize, Diagnose & Recover from Errors
 - Error messages are missing
 - Calculation errors are not explained
 - Logs contain technical jargon
 - Users don't include enough data to understand issues
10. Help & Documentation
 - New users require training → lack of onboarding, help, or guidance

personas

persona 1 (Nehal)

- Name:** Nehal
Role: Student Information System (SIS) Admin
Experience: 3+ years working with student data & academic records
Tech Comfort: Moderate knows her way around systems but hates over complicated interfaces
Work Environment: Fast paced office, constantly multitasking, dealing with student records, grade updates, and reports
About Her
Nehal manages student information every single day. She handles transcripts, edits student data, checks grades, and supports faculty who rely on accurate, up to date info. She needs a system that's fast, clear, and doesn't make her find a problem with it.
- Goals**
- Access student transcripts instantly
 - Navigate the SIS quickly without clicking 1000 random menus
 - Update student data with fewer steps
 - Ensure grades (pass/fail) are visible and easy to manage
 - Reduce confusion caused by inaccurate or unclear naming
 - Work smoothly from mobile when she's not at her desk
- Frustrations**
- Transcripts are hidden behind multiple sections
 - Navigation feels confusing instead of helpful
 - Important data (like pass/fail grades) is not easy to find
 - Editing student info requires too many manual steps
 - System naming is inconsistent, causing mistakes
 - Mobile experience is tragic not friendly at all
- Motivations**
- Wants efficiency less clicking, more doing
 - Wants reliability and accuracy
 - Wants an interface that matches her workflow, not fights against it
 - Wants a mobile version/app that actually works

Problem Statement

Nehal, a SIS admin, struggles to perform her daily tasks efficiently because the current backend system is unintuitive, poorly structured, and requires excessive manual steps. Critical information like transcripts and passing/failing grades is hidden or difficult to access, leading to delays and potential errors. Inconsistent naming causes confusion, and the mobile version is not functional enough for quick on the go tasks. Nehal needs a simplified, clearly organized, and more automated SIS that supports fast navigation, accurate data management, and a usable mobile experience.

persona 2 (Asmaa)

- Name:** Asmaa
Role: IT Specialist (SIS Technical Support)
Experience: 3+ years supporting and maintaining institutional systems
Environment: Works with faculty, admins, and technical teams to ensure the SIS runs smoothly
About Her
Asmaa is the go to person whenever the system acts up. She handles technical issues, monitors errors, checks calculations, and deals with data conflicts. When something breaks, everyone calls her. She needs a system that's predictable, transparent, and safe.
- Goals**
- Ensure accuracy in system calculations
 - Get full visibility of important student or system information
 - Prevent accidental data overwrites
 - Receive clear error messages so she can fix issues fast
 - Reduce the time spent troubleshooting preventable problems
 - Improve system stability and data reliability
- Frustrations**
- Critical information is hidden or hard to access
 - Data can be overridden with ZERO warning
 - System calculations sometimes glitch or produce wrong results
 - Almost no error messages, she's blind until something breaks
 - Error prevention is weak, leading to unnecessary technical issues
 - She wastes time fixing problems the system should've stopped
- Motivations**
- Wants a reliable, stable backend
 - Wants transparency in system actions
 - Wants better validation and safeguards
 - Wants tools to diagnose issues easily
 - Wants the SIS to require LESS "manual monitoring" from IT

Problem Statement

Asmaa, an IT specialist responsible for maintaining the SIS, struggles to ensure system reliability because critical information is not clearly visible, data can be overwritten without warning, and system calculations often lack accuracy. The current SIS provides little to no error messaging or prevention, making it difficult for her to detect issues early or troubleshoot efficiently. She needs a more transparent, error resistant system with accurate calculations, visible alerts, and strong validation to reduce system failures and technical risks.

persona 3 (Mohammed)

- Name:** Mohammed A.
Role: Admissions & Student Records Officer
Experience: 3+ years working with SIS and academic admin systems
Tech level: Medium comfortable but not advanced
Goals:
- Process student admissions smoothly
 - Get accurate data fast
 - Approve courses + manage advisors without mistakes
 - Generate reports easily without manual work
- Responsibilities:**
- Filtering applicants
 - Accepting/denying courses
 - Assigning academic advisors
 - Managing student actions (transfers, specialization changes, etc.)
 - Preparing reports for departments
- Frustrations / Pain Points:**
- Filtration is inaccurate + doesn't allow multiple filters
 - Course acceptance flow is confusing and hidden in menus
 - Student actions are slow and require too many steps
 - Advisor assignment is one by one, super time consuming
 - Course tags are unclear and don't show enough detail
 - Reports must be exported manually takes hours
 - Similar-looking pages cause wrong actions
 - No notifications when something changes
 - Logs are vague and full of technical jargon
 - New employees struggle because the user flow is complicated
- Needs:**
- Accurate and flexible filtering
 - Clear student info at a glance
 - Faster workflows with fewer clicks
 - Automated report generation
 - Clear UI differences between different tasks
 - Meaningful action logs
 - Notifications + warnings for changes

Problem Statement

Mohammed, an admissions officer, struggles to complete essential tasks in the SIS because the system's filtration is inaccurate, workflows are complicated, and important information is hidden across multiple menus. Assigning advisors, accepting courses, and performing student actions require repetitive, manual steps that slow down his work and lead to errors. Reports must be created manually, the interface looks almost the same across different tasks, and the system provides no warnings or meaningful logs making it hard to track changes or onboard new staff. Admissions staff need a clearer, more automated, and more intuitive SIS that allows accurate filtering, supports batch actions, generates reports automatically, and provides transparent notifications and logs so they can manage student data confidently, quickly, and without risk of mistakes.

overall prob and solutions

OVERALL PROBLEMS IN THE CURRENT SIS (From Admin + IT Perspectives)

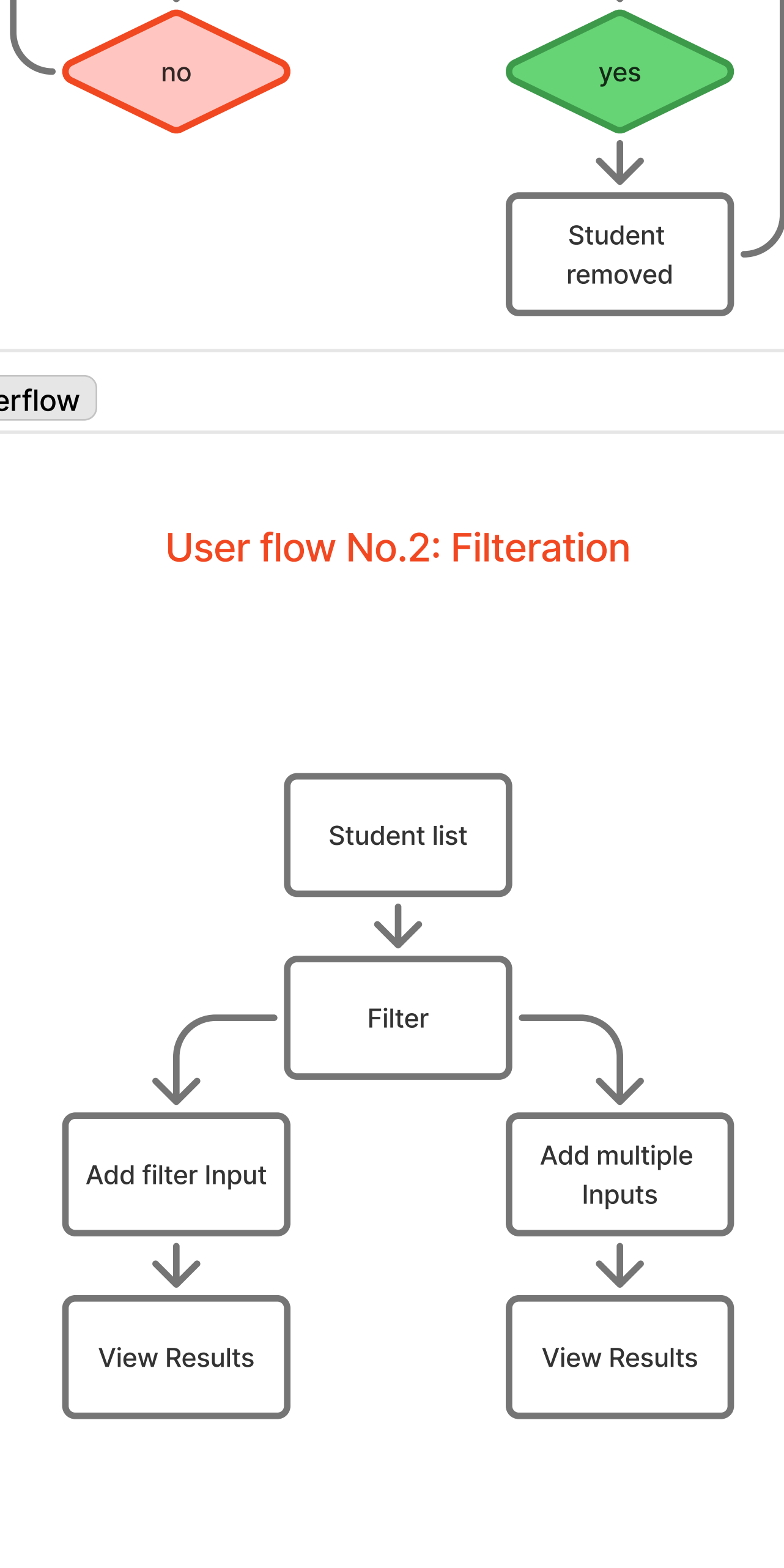
1. **Important information is hard to find**
 - Transcripts aren't visible directly
 - Critical data (grades, status, warnings) is buried
 - IT says important backend info is not visible either
2. **Navigation is confusing + not intuitive**
 - Admin struggles to find things easily
 - Structure isn't logical, causes mistakes
 - Mobile version is basically unusable
3. **Data entry + editing is outdated and time-consuming**
 - Changing student data requires multiple steps
 - No automation
 - Information can be overwritten without any warning
4. **System lacks accuracy and reliability**
 - SIS can show wrong calculations
 - Passing/failing status isn't clear
 - Naming and labels are inaccurate, leads to confusion
5. **Weak error prevention + no error messages**
 - No warnings before overriding data
 - No error messages when something breaks
 - No validation checks to prevent mistakes
 - IT spends time fixing issues that should've been prevented
6. **Mobile experience is terrible**
 - Not intuitive
 - Not designed for on the go admin/IT usage
 - Should be a proper app
7. **Communication inside the system is unclear**
 - Users don't know what the system is doing
 - No transparent logs or alerts
 - No feedback from important actions

SOLUTIONS (UX + System Improvements)

1. **Make key information visible immediately**
 - Dashboard showing: transcripts, grades, warnings, student status
 - Clear alerts for missing or incorrect data
 - Quick access panels for frequently used features
2. **Redesign navigation to be logical and role based**
 - Group features by task, not by system structure
 - Use a clear left sidebar or tabbed navigation
 - Add a search bar for fast access to students, courses, records
 - Make mobile layout responsive or build a dedicated mobile app
3. **Streamline data editing workflow**
 - One page form for student profile editing
 - Auto fill + automation for repetitive fields
 - Add step by step wizards for complex actions
 - Save progress + show what was changed
4. **Improve system accuracy with validation + auto calculation checks**
 - Add verification steps before finalizing grades
 - Show calculation logs when generating GPAs or averages
 - Standardize naming across the entire system
 - Add logic rules that prevent impossible data combos
5. **Add strong error handling + prevention**
 - Warning pop ups before overriding data
 - Real time validation (e.g., "this field cannot be empty")
 - Clear error messages that explain what went wrong
 - Logs for all important actions transparent for IT
6. **Build a better mobile experience**
 - Fully responsive design
 - Simple navigation on mobile
 - Or: Create a compact admin mobile app for quick actions
 - Keep only the most essential features on mobile
7. **Add system feedback, transparency, and communication**
 - Success confirmations ("student data saved successfully")
 - Visual indicators for errors, warnings, required fields
 - Change history for each student record
 - Notifications for conflicting data updates

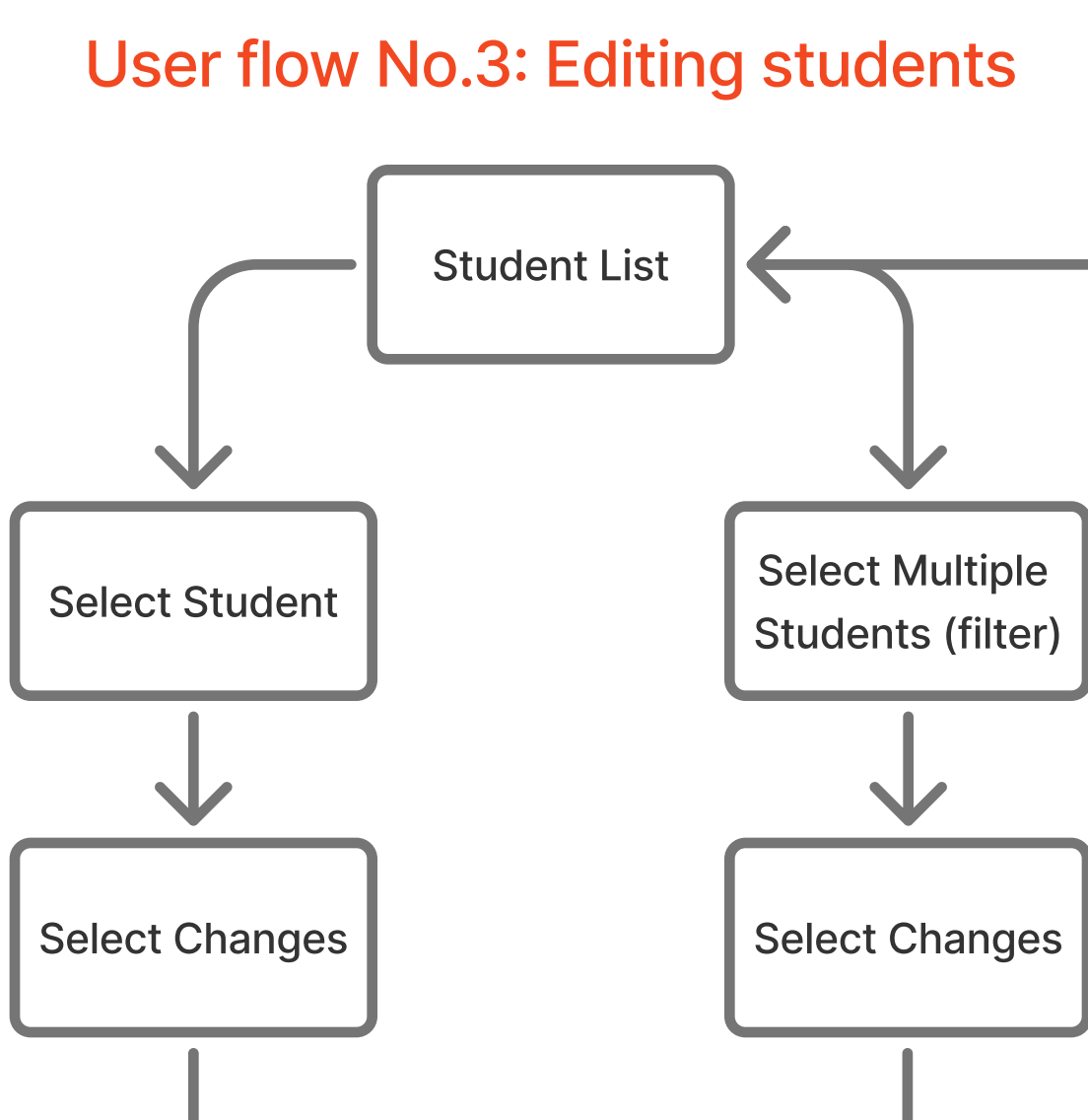
Userflow

User flow No.1: Error Prevention



Userflow

User flow No.2: Filtration



Userflow

User flow No.3: Editing students

